

Scenario Demonstration 2: Data Analysis

Independent Task Sheet | Phase: Actual | SD2

Organisation: Apex Sports Academy | **Analyst:** Jamie | **Period:** January – December 2024

Business Question

Which session types, facilities and membership tiers generate the highest bookings and attendance rates, and what patterns suggest members are at risk of disengaging?

Tasks

1. Summary Statistics

- Calculate: total bookings, attended count, no-show count, cancelled count, unknown count.
- Verify all status counts sum to total bookings.
- Calculate total revenue (attended bookings only), average fee, min fee, max fee.
- Calculate the overall attendance rate as a percentage.

2. Conditional Analysis

- Build a session type comparison table (Boxing, Pilates, Spin, Swim, Yoga).
- Include: total bookings, attended, no-show, attendance rate, total revenue per session type.
- Build a membership tier comparison table (Bronze, Silver, Gold, Platinum).
- Identify which session type and which membership tier performs best on each metric.

3. Validation and Anomaly Review

- Cross-check: confirm formula totals reconcile with raw data row counts.
- Apply conditional formatting to flag suspicious values (e.g. zero fees on attended bookings).
- Identify and document any anomalies. State whether each is an error or a real signal.
- Write a confidence statement for the data and note known limitations.

4. Pivot Tables, Charts and Slicers

- Create at least two Pivot Tables. Verify grand totals match raw data counts.
- Create at least one Pivot Chart with title, axis labels and connected slicer.
- Add a timeline filter connected to at least one pivot.

5. Trendline Analysis

- Create a monthly bookings summary table (month number 1-12 vs. total bookings).
- Insert a Scatter chart. Add a linear trendline.
- Display the equation ($y = mx + c$) and R-squared value on the chart.
- Write a 2-3 sentence interpretation of the trendline and R-squared.

6. Dashboard and Written Interpretation

- Create a Dashboard sheet with: 3 summary cards, 1 Pivot Chart, 1 slicer, 1 timeline.
- Write a full interpretation: what the data shows, what it suggests, what it does not prove.

- Include one specific recommendation for Priya and Marcus.
- Note what further data would improve confidence in the findings.

Required Deliverables

• Summary_Stats sheet — all core metrics calculated and labelled.
• Session_Analysis and Membership_Analysis sheets — COUNTIFS/SUMIFS tables.
• Validation sheet — cross-checks, anomaly log, confidence statement.
• At least 2 Pivot Tables with verified totals.
• Pivot Chart with slicer and timeline.
• Scatter chart with trendline, equation and R-squared.
• Dashboard sheet — summary cards, chart, slicer, timeline.
• Written interpretation (minimum 4 sentences).